

Satyam Awasthi

Researcher & Software Engineer

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Research Summary

Researcher working at the intersection of **neural interfaces**, **physiological sensing**, **learning**, and **control**. Current work spans **EEG-based brain-computer interfaces** and **robust output-feedback model predictive control**.

Research Interests

Neural Interfaces, Brain-Computer Interfaces, Physiological & Wearable Sensing, Human-Machine Interaction, Machine Learning for Biosignals, Robust State Estimation, and Model Predictive Control.

Research Experience

Autonomous Systems Group, Imperial College London

Research Collaboration

Jul 2026 – Present

Advisor: [Dr. Johannes Köhler](#)

- Investigating **state-dependent estimation and tracking-error bounds** for nonlinear robust output-feedback MPC with state-dependent process and measurement uncertainty.
- Developing coupled observer and tracking-error tube propagation using **contraction-based certificates** for robust constraint tightening and uncertainty-aware trajectory planning.

Four Eyes Laboratory, University of California, Santa Barbara

Graduate Researcher

Sep 2021 – Jun 2023

Advisor: [Prof. Tobias Höllerer](#); Co-Advisor: [Prof. Michael Beyeler](#)

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Investigated how **oculomotor behaviors—including fixation, saccades, and smooth pursuit**—affected gaze-tracking precision and temporal response, including interactions with device-side signal filtering during locomotion.
- Designed human-subject experiments and analyzed gaze-tracking accuracy, calibration drift, and motion-dependent sensing error across multiple mixed-reality devices.
- Released a reproducible **calibration framework** and **multi-device eye-tracking dataset** containing gaze trajectories, calibration measurements, and locomotion data.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur

Undergraduate Researcher

Jan 2016 – Dec 2018

- Contributed perception, multimodal state-estimation, and navigation components to **Eklavya 6.0**, IIT Kharagpur's autonomous ground vehicle that finished **Runner-Up at IGVC 2018**.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

COMSNETS, 2022.

DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

IEEE VR Workshops (VRW), 2024.

DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

IEEE ISMAR Adjunct, 2023.

DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Selected Research Projects

EEG-Based BCI for 3D Navigation

2026 – Present

Independent Brain-Computer Interface Project

- Developing a **4-channel EEG acquisition and decoding pipeline** for real-time control of an interactive 3D navigation environment.
- Building an initial **SSVEP-based intent-decoding system** using neural-signal preprocessing and spectral / correlation-based features for discrete movement commands.
- Investigating confidence thresholds and temporal evidence accumulation to characterize the trade-off between **decoding reliability and command latency**.

SmartWatch for Wall Writing

2019 – 2022

Wearable Inertial Sensing for Human-Computer Interaction

- Contributed to a wearable sensing system for **real-time transcription of free-form wall writing** from smartwatch inertial measurements.
- Explored the use of human wrist-motion signals as an unobtrusive input modality for recognizing physical writing gestures.

JitterNot

Jan 2022 – Mar 2022

Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput to support closed-loop decisions under non-stationary network dynamics.
- Designed a receding-horizon **model predictive control (MPC)** framework for adaptive bitrate selection under bandwidth constraints.

In Motion — Wearable Human Activity Recognition

Nov 2019

DeepConvLSTM Modeling of Inertial Sensor Signals (IIT Kharagpur)

- Designed a streaming inference pipeline for **high-frequency IMU signals** enabling temporal human-activity classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with recurrent temporal modeling of wearable sensor sequences.

Education

University of California, Santa Barbara (UCSB)

Santa Barbara, CA

M.S. in Computer Science (GPA: 3.91/4.00)

Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

Kharagpur, India

B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Jul 2016 – Jul 2020

Industry Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agent pipelines** for intent routing, structured tool execution, and generative user-interface composition.
- Developed production platform infrastructure supporting large-scale client applications and internal developer workflows.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed production Android systems for Yahoo Mail, including data-driven inbox commerce and expense-management features.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Developed user-facing search and product-discovery features for Coupang's Android commerce platform.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Developed Android platform and device applications involving media processing, privacy, and isolated user-storage architectures.

Teaching

University of California, Santa Barbara

Graduate Teaching Assistant

Santa Barbara, CA

Sep 2021 – Jun 2023

- Led instructional labs, developed programming assignments, and supported senior capstone teams across CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, MATLAB, Java, Kotlin

Machine Learning & Signal Processing: PyTorch, NumPy/SciPy, OpenCV, Time-Series Modeling, Spectral Analysis, Digital Filtering, Wearable/Inertial Signal Processing

Control & Estimation: Model Predictive Control (MPC), Robust MPC, Contraction-Based Methods, State Estimation (EKF), Sensor Fusion, Motion Planning

Robotics & Interactive Systems: ROS/ROS2, Gazebo, RViz, Unity, OpenGL

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024

ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
STATEMENT OF GRADES OBTAINED FOR THE 8 SEMESTER COURSE IN ENGINEERING/TECHNOLOGY LEADING TO THE AWARD OF
BACHELOR OF TECHNOLOGY (HONOURS)



Roll No: 16EE10042

Name: SATYAM AWASTHI

Year of Admission : 2016-2017

Year of Graduation : 2019-2020

Course: B.Tech.(Hons.) in ELECTRICAL ENGINEERING

For Semester 1		SGPA: 8.09	CGPA: 8.09		
Subno	Name	L-T-P	CRD	GRD	
CY11001	CHEMISTRY	3-1-0	4	B	
CY19001	CHEMISTRY LAB.	0-0-3	2	EX	
EA10001	EXTRA ACADEMIC ACTIVITY-I	0-0-3	0	B	
EE11001	ELECTRICAL TECHNOLOGY	3-1-0	4	C	
EE19001	ELECTRICAL TECHNOLOGY LAB.	0-0-3	2	B	
HS13001	ENGLISH FOR COMMUNICATION	3-0-2	4	B	
MA10001	MATHEMATICS-I	3-1-0	4	B	
ME19001	INTRODUCTION TO MANUFACTURING PROCESSES	0-0-3	2	A	

For Semester 2		SGPA: 8.91	CGPA: 8.51		
Subno	Name	L-T-P	CRD	GRD	
CE13001	ENGINEERING DRAWING AND COMPUTER GRAPHICS	1-0-3	3	A	
CS10001	PROGRAMMING AND DATA STRUCTURES	3-0-0	3	EX	
CS19101	PROGRAMMING AND DATA STRUCTURES TUTORIAL AND LABORATORY	0-1-3	3	EX	
EA10002	EXTRA ACADEMIC ACTIVITY-II	0-0-3	0	B	
MA10002	MATHEMATICS-II	3-1-0	4	A	
ME10001	MECHANICS	3-1-0	4	B	
PH11001	PHYSICS	3-1-0	4	B	
PH19001	PHYSICS LAB.	0-0-3	2	A	

For Semester 3		SGPA: 8.68	CGPA: 8.57		
Subno	Name	L-T-P	CRD	GRD	
EA10003	EXTRA ACADEMIC ACTIVITY-III	0-0-3	0	Y	
EC21103	INTRODUCTION TO ELECTRONICS	3-1-0	4	B	
EC29003	INTRODUCTION TO ELECTRONICS LAB.	0-0-3	2	B	
EE21101	SIGNALS & NETWORKS	3-1-0	4	B	
EE29001	SIGNALS & NETWORKS LAB.	0-0-3	2	A	
HS30103	ECONOMICS OF HEALTH	3-0-0	3	EX	
MA20101	TRANSFORM CALCULUS	3-0-0	3	A	
PH20003	PHYSICS-II	3-1-0	4	A	

For Semester 4		SGPA: 8.96	CGPA: 8.67		
Subno	Name	L-T-P	CRD	GRD	
BS20001	SCIENCE OF LIVING SYSTEM	2-0-0	2	A	
EA10004	EXTRA ACADEMIC ACTIVITY-IV	0-0-3	0	Y	
EC21008	ANALOG ELECTRONIC CIRCUITS	3-1-0	4	A	
EC29008	ANALOG CIRCUITS LAB.	0-0-3	2	EX	
EE21002	ELECTRICAL MACHINES	3-1-0	4	A	
EE21004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS	3-1-0	4	A	
EE29002	ELECTRICAL MACHINES LAB.	0-0-3	2	A	
EE29004	MEASUREMENTS AND ELECTRONIC INSTRUMENTS LAB.	0-0-3	2	EX	
EV20001	ENVIRONMENTAL SCIENCE	2-0-0	2	B	
HS30074	SCIENCE AND HUMANISM	3-0-0	3	B	

For Semester 5		SGPA: 9.04	CGPA: 8.75		
Subno	Name	L-T-P	CRD	GRD	
CS40019	IMAGE PROCESSING	3-0-0	3	A	
EC31003(*)	DIGITAL ELECTRONIC CIRCUITS	3-1-0	4	A	
EC39003(*)	DIGITAL ELECTRONIC CIRCUITS LAB.	0-0-3	2	EX	
EE31009	CONTROL SYSTEM ENGINEERING	3-1-0	4	A	
EE31011	POWER ELECTRONICS	3-1-0	4	B	
EE39006	POWER ELECTRONICS LAB.	0-0-3	2	EX	
EE39009	CONTROL AND INSTRUMENTATION LAB.	0-0-3	2	B	
EP60025	SPECIAL TOPICS IN ENTREPRENEURSHIP	2-1-0	3	EX	

For Semester 6		SGPA: 9.46	CGPA: 8.87		
Subno	Name	L-T-P	CRD	GRD	
CS31702(*)	COMPUTER ARCHITECTURE AND OPERATING SYSTEM	4-0-0	4	EX	
CS60092	INFORMATION RETRIEVAL	3-0-0	3	B	
EE30004	EMBEDDED SYSTEMS	3-0-0	3	EX	
EE30012	ELECTROMAGNETIC THEORY & APPL.	3-0-0	3	A	
EE31002	POWER SYSTEMS	3-1-0	4	A	
EE39002	POWER SYSTEMS LAB.	0-0-3	2	EX	
EE39004	EMBEDDED SYSTEMS LAB.	0-0-3	2	EX	
MA20106	PROBABILITY & STOCHASTIC PROCESSES	3-0-0	3	EX	

For Semester 7		SGPA: 9.25	CGPA: 8.92		
Subno	Name	L-T-P	CRD	GRD	
CS60009(*)	SMARTPHONE COMPUTING & APPLICATION	3-0-0	3	EX	
CS60021	SCALABLE DATA MINING	3-0-0	3	A	
EE41001	INDUSTRIAL INSTRUMENTATION	3-1-0	4	B	
EE47009	PROJECT-I	0-0-0	3	EX	
EE48001	INDUSTRIAL TRAINING	0-0-0	2	A	
EE49001	CONTROL AND ELECTRONIC SYSTEM DESIGN	0-0-3	2	A	
EP60044	FRUGAL ENGINEERING	3-0-0	3	EX	

For Semester 8		SGPA: 9.63	CGPA: 8.98		
Subno	Name	L-T-P	CRD	GRD	
CS31006	COMPUTER NETWORKS	3-0-0	3	EX	
EE40002	ELECTRIC DRIVES	3-0-0	3	B	
EE47008	PROJECT - PART 2	0-0-0	6	EX	
EE48002	COMPREHENSIVE VIVA VOCE	0-0-0	2	EX	
EE49004	POWER APPARATUS & SYSTEM DESIGN	0-0-3	2	EX	

Additional subjects taken into account for earning a Minor		L-T-P	CRD	Semno	GRD
Subno	Name				
CS21003	ALGORITHMS - I	3-1-0	4	4	A
CS29003	ALGORITHMS LABORATORY	0-0-3	2	4	A
CS60050	MACHINE LEARNING	3-0-0	3	5	B
CS60118	CLOUD COMPUTING	3-0-0	3	8	EX

* sign against a major curricular subject indicates that it has been taken into account for Minor

GPA in Minor: 9.36

Minor in : COMPUTER SCIENCE AND ENGINEERING

Details of other additional subjects

Subno	Name	L-T-P	CRD	Semno	GRD
EC21107	SEMICONDUCTOR DEVICES	3-1-0	4	3	A

Total Additional Credit Taken: 16
GPA in Additional Subjects: 9.00

Total Additional Credit Cleared: 16

Total Credit Taken in Major Curriculum: 176 Total Credit Cleared: 176 CGPA: 8.98

Checked by

Joint Registrar (Academic)

Joint Registrar (Academic) / Assistant Registrar(UG)

Date of Issue : 16 Jul 2020

University of California, Santa Barbara

Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPVIN

PRINTED: 04/05/23

PAGE: 1

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI

PERM NUMBER: 807498

SSN: XXX-XX-4913

Birth Date: 01/23/XX

CURRENT ACADEMIC PROGRAM:

COLLEGE: College of Engineering

OBJECTIVE: Master of Science

MAJOR: Computer Science

COURSE	TITLE	GRADE	COMPLETE UNITS	GPA UNITS	GRADE POINTS	GPA
2021 Fall Quarter						
Admitted into the Master of Science Program in Computer Science						
Admitted into the Master of Science Program in Computer Science						
CMPSC	291A	SPEC TOPIC APPS GEN	A+	4.0	4.0	16.00
CMPSC	501	TECH COMP SCI TEACH	S	1.0	0.0	0.00
CMPSC	502	TEACHING COMPTN SCI	S	3.0	0.0	0.00
LING	4	INDIV & GRP INSTRU	P	0.0	0.0	0.00
No Baccalaureate Credit Lower-Division Course						
Graduate Total:				8.0	4.0	16.00
THRU F21				8.0	4.0	16.00
2022 Winter Quarter						
CMPSC	271	ADV DIST SYS	A	4.0	4.0	16.00
CMPSC	291A	SPEC TOPIC APPS GEN	A	4.0	4.0	16.00
CMPSC	293C	SPEC TOP SYS PROG L	A	4.0	4.0	16.00
Graduate Total:				12.0	12.0	48.00
THRU W22				20.0	16.0	64.00
2022 Spring Quarter						
CMPSC	292C	SPEC TOP FOUN PROG	A+	4.0	4.0	16.00
CMPSC	293N	SPEC TOPICS CS SYS	A	4.0	4.0	16.00
CMPSC	595C	PRG LANG SOFT ENGR	S	2.0	0.0	0.00
Graduate Total:				10.0	8.0	32.00
THRU S22				30.0	24.0	96.00
2022 Summer Session						
CMPSC	193	INTERN IN INDUSTRY	P	1.0	0.0	0.00
Graduate Total:				1.0	0.0	0.00
THRU M22				31.0	24.0	96.00

CONTINUED TO PAGE 2

ISSUED TO:

SATYAM AWASTHI
6510 El Colegio Rd Apt 1102
Santa Barbara CA 93106-4200
Anthony Schmid
University Registrar

University of California, Santa Barbara

Office of the Registrar, Santa Barbara, CA 93106-2015

PARCHMENT: TWCTPV1N

PRINTED: 04/05/23

PAGE: 2

OFFICIAL TRANSCRIPT

STUDENT NAME: SATYAM AWASTHI

PERM NUMBER: 807498

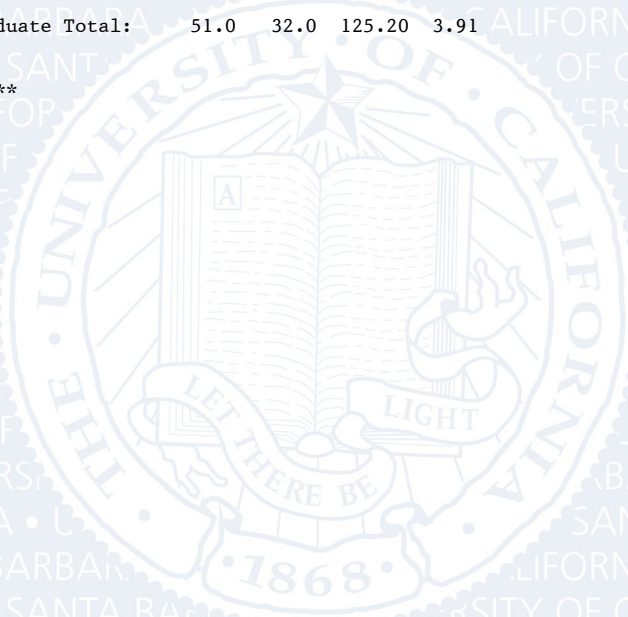
			COMPLETE GPA		GRADE POINTS		GPA
COURSE		TITLE	GRADE	UNITS	UNITS		

2022 Fall Quarter							
CMPSC	263	RUNTIME SYSTEMS	A	4.0	4.0	16.00	
CMPSC	291A	SPEC TOPIC APPS GEN	B+	4.0	4.0	13.20	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
Graduate Total:				12.0	8.0	29.20	3.65
THRU F22	Graduate Total:			43.0	32.0	125.20	3.91

2023 Winter Quarter							
CMPSC	502	TEACHING COMPTR SCI	S	4.0	0.0	0.00	
CMPSC	596	DIRECTED RESEARCH	S	4.0	0.0	0.00	
Graduate Total:				8.0	0.0	0.00	0.00
THRU W23	Graduate Total:			51.0	32.0	125.20	3.91

UC Credit	Graduate Total:			51.0	32.0	125.20	3.91

** END OF RECORD **



ISSUED TO:

SATYAM AWASTHI

6510 El Colegio Rd Apt 1102

Santa Barbara CA 93106-4200



Anthony Schmid

University Registrar

UNIVERSITY OF CALIFORNIA • SANTA BARBARA

TRANSCRIPT GUIDE

UNITS OF CREDIT:

Until September 1966, credits were recorded as semester units. Since that time, the University of California, Santa Barbara has operated on a quarter calendar, with three quarters per academic year and an optional summer session. Each quarter has ten weeks of instruction at the rate of one unit for every three hours of student work required each week.

CURRENT GRADING SYSTEM:

The grades A, B, C, and D may be modified by plus (+) or minus (-) suffixes. Grade points for each unit are assigned as follows:

A+ = 4.0	A = 4.0	A- = 3.7
B+ = 3.3	B = 3.0	B- = 2.7
C+ = 2.3	C = 2.0	C- = 1.7
D+ = 1.3	D = 1.0	D- = 0.7

Grades of F, I, IP, P, NP, S, U and W = 0.0 grade points

Unit credit, but not grade-point credit, is assigned for P and S grades.

P = Passed (C or better)	Undergraduate Course Only
NP = Not Passed (C- or below)	Undergraduate Course Only
S = Satisfactory (B or better)	Graduate Course Only
U = Unsatisfactory (B- or below)	Graduate Course Only
I = Incomplete	
W = Withdrawal after established deadline	
IP = Grade to be assigned at the end of the final course in the sequence	

GOOD ACADEMIC STANDING:

A student is in good standing academically unless otherwise indicated.

TRANSFER CREDIT:

Only credit that is accepted by the University is shown on the transcripts of UCSB students. Credit from other UC campuses may be shown either by individual course records or as a unit summary by campus. Individual courses from schools outside of the UC system are not shown.

ADVANCED PLACEMENT CREDIT:

Examinations and the credits accepted for them are shown on the transcript in the same format and location as transfer credit.

COURSE NUMBERING SYSTEM:

1 - 99	Lower Division Courses
100 - 199	Upper Division Courses
200 - 299	Graduate Courses
300 - 399	Professional courses for teachers
400 & above	Other professional and graduate courses

REPEATED COURSES:

An undergraduate student may repeat only those courses for which an NP or letter grade of C- or lower is recorded. If the grade from a previous attempt had been included in the GPA calculations, its effect is then removed by a "repeat adjustment" up to a maximum of 16 units over the student's career.

ACCREDITATION:

Western Association of Schools and Colleges (WASC)